

# Cell-State Reachability

MEASURED → MODELLED → TEST NEXT

A screen-relative test of transcriptomic direction — not a state-conversion claim

1

Inputs

source objects reused from Zhu workflow

TARGET DIRECTION **d**

Source-study-reused population contrast (not independent or a trajectory)

EFFECT DICTIONARY **E**

Screen-derived CRISPRi DE z-scores in post-expansion Rest CD4 cells

Target union: 25,672; shared screen: 8,950

Final merged analysis: 6,188 genes

Dictionary is donor-collapsed; not measured in polarized Th2 cells

2

Projection

what the model computes

target

unmatched residual

projected model fit

Non-negative linear combinations in declared effect space

Residual is model-relative

3

Challenge

source-bound + measured transfer tests

0.444 ± 0.018

mean ± SD, 12 hash-frozen gene splits

selected population contrast; not trajectory

range: 0.417-0.473

Frozen transfer challenges

Source gain +0.087 / +0.090

Donor-pair gain +0.032; nRMSE 1.153 vs 1.018

Arce ranks: 0.148/0.084/0.088; donor ranks: 0.73-0.93; signs: 50%-64%

Zhu arrayed RNA: 9/9 retrieved; centered cosine 0.580

RNA→flow cytokine ranks: 0.72-0.85; descriptive, not donor/state validation

4

Decide

what this result changes

REPLICATION NEEDED

Donor-resolved, polarized Th2 starting-state study

MEASURE NEXT

Matched modalities, combinations + functional readouts

NOT ESTABLISHED

State conversion, rescue, or target validation

## CLAIM BOUNDARY

Screen-relative directional support • measured subset only • model-relative geometry ≠ biological efficacy